Rock Climbing Robots

Hackathon Rulebook

Rules revision **v1.1** — October 20, 2026

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1 Overview

Rock Climbing Robots is a hackathon in which teams design, build and program small legged robots that climb an instrumented artificial climbing wall. Each robot is secured by a safety rope running over a pulley at the top of the wall, exactly as a human climber would be.

Website: `https://rock-climbing-robots.github.io/`

The event combines mechanical design (limbs, grippers, anchor point), control (balance and contact planning on a sparse set of holds) and perception (two external cameras observe the wall).

- **Teams:** 2 to 6 members per team.

- **Deliverable:** a robot able to climb from the start zone to the top of the wall using the mounted holds.
- **Scoring:** two independent evaluations, in simulation and on the real robot (Section 8).

2 Robots

2.1 Morphology

The robot climbs using its *limbs*; its shape is otherwise not constrained. Other parts of the robot, such as its trunk, may also touch the wall and the holds.

A *humanoid* robot has **four limbs**: a central trunk carrying two upper limbs and two lower limbs. Whether a robot is humanoid is left to the discretion of the organisers. Non-humanoid robots are allowed, with a time factor (Section 8).

Attachment to the wall must be made through the holds.

2.2 Size

The maximum height of the robot is **400 mm**, measured with the robot standing in its nominal upright posture, arms along the body.

Limbs may extend beyond that envelope while climbing; the 400 mm limit applies to the reference posture only.

There is **no limit** on the width or the depth of the robot.

The mass of the robot must not exceed **2 kg**.

2.3 Power

A robot may be powered either:

(a) by **on-board batteries**, or

(b) by an **external power supply**, whose cable is routed along the safety rope down to the robot (see Section 5).

A tethered cable must be attached to the safety rope and must never be load-bearing: it shall not be able to hold the robot if the anchor point fails, and it shall not be used to help the robot climb.

2.4 Control and autonomy

Computation may be on-board or off-board. Communication with external computers is free: the Ethernet tether or any wireless link (Wi-Fi, Bluetooth, etc.), on any band, may be used.

The robot must be **autonomous**: **teleoperation is not allowed**.

2.5 Provided robots

Robots will be provided by the organisers to teams that do not bring their own platform.

The provided platform is derived from **Microban**, an open-source, fully 3D-printable humanoid robot of about 30 cm developed by the Rhoban team (LaBRI, University of Bordeaux), with **limbs adapted for climbing**.

Item	Value
Actuators	19× Dynamixel `XL330-M288-T`
Degrees of freedom	6 per leg, 3 per arm
Height	≈ 300 mm (base Microban)
Repository	`https://github.com/Rhoban/microban`

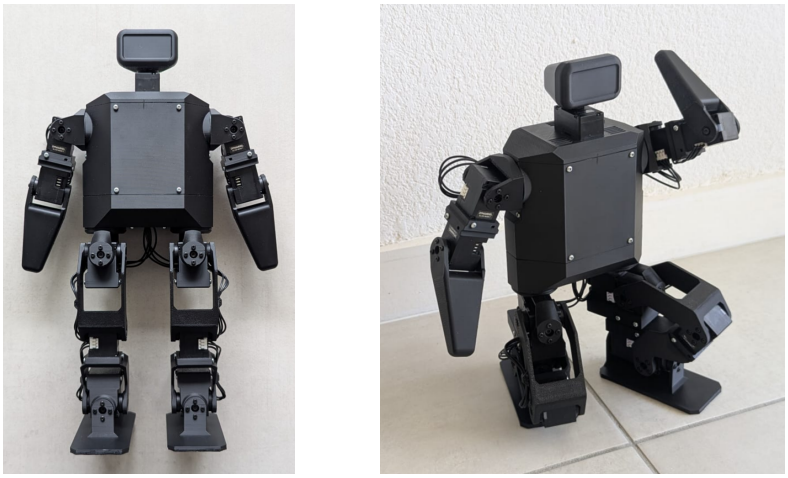


Figure 1: **Microban**, the base platform for the provided robots, shown here in its standard configuration — the climbing version uses adapted limbs. Photographs from the Microban repository (Rhoban team, LaBRI, University of Bordeaux).

3 Climbing wall

Four walls are provided by the organisers. The boards are identical: the difficulty comes from the configuration of the holds, which are reconfigured during the event. There are **three difficulty levels**:

- **Level 1** — a "ladder": regularly spaced holds;
- **Level 2** — a ladder with missing holds;
- **Level 3** — the robot needs to move to the right or to the left.

Each climbing wall is a flat rectangular board, mounted vertically, of **1400 mm** height and **800 mm** width.

The board is made of **plywood**, with a thickness of **10 mm**, mounted on a rigid frame.

The board is drilled with a regular square grid of **M3** holes with a pitch of **15 mm**. The border margin is 10 mm on all four sides. Holes are plain through-holes: holds are bolted from the front and tightened by nuts on the rear side of the board.

4 Holds

Holds compatible with the climbing variant of the Microban platform are provided by the organisers.

Teams may come with **their own holds**, or adapt the provided ones, as long as they remain compatible with the wall format (Figure 3):

- a hold is fixed with **two M3 bolts**, on **two holes side by side horizontally** on the grid, i.e. **15 mm apart**; the bolts go through the hold and the board, and are tightened by **nuts**

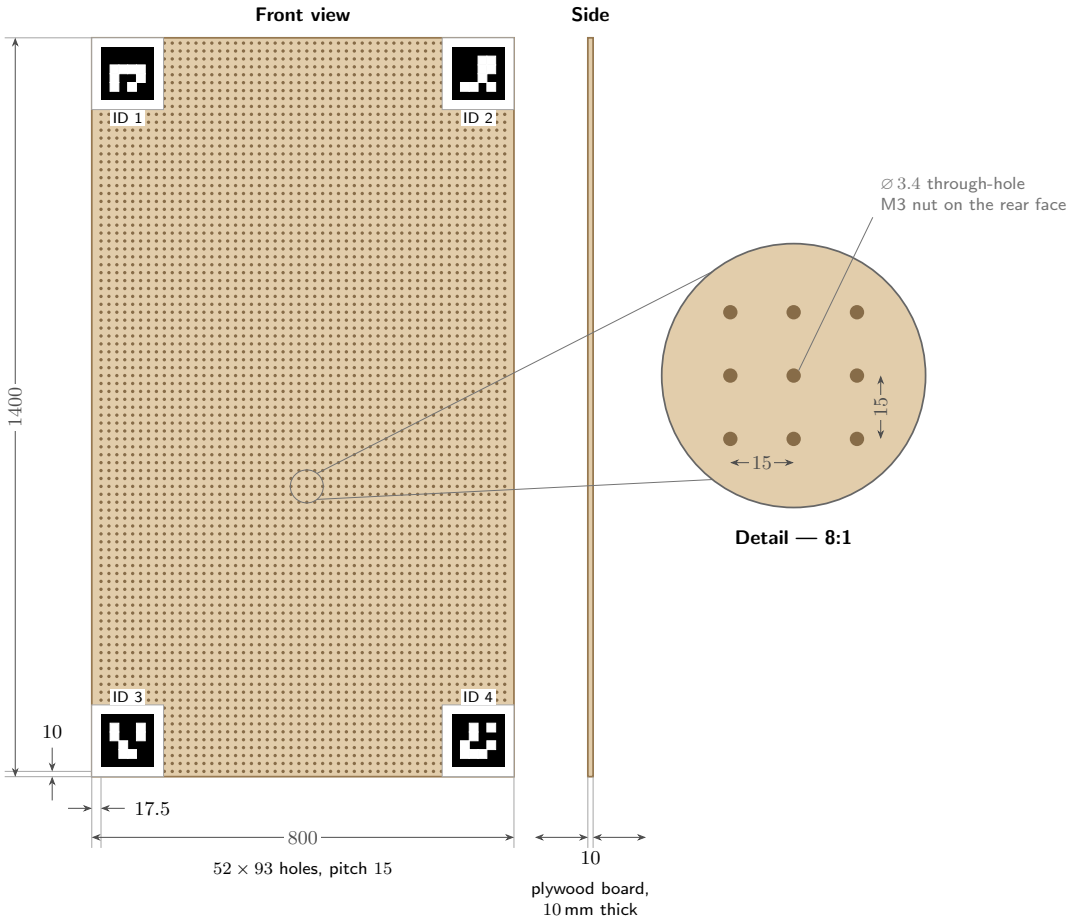


Figure 2: Climbing wall — front and side views. All dimensions in millimetres. Hole positions are shown enlarged for readability.

on the opposite side of the wall;

- a hold must fit inside a **cylinder of diameter 60 mm (radius 30 mm) and length 100 mm**, whose axis is normal to the wall and centred between the two bolts.

5 Setup

The complete setup around one wall is shown in Figure 4.

5.1 Cameras

Two external cameras are available per wall, one on each side, connected over **USB**. They are provided and positioned by the organisers.

It is up to the teams to **plug the USB cables** into their own computer and to **retrieve the images**. Teams must only use the **RGB components** of the camera streams.

Both cameras observe the full wall surface. Their images are available to the teams for perception and for run recording.

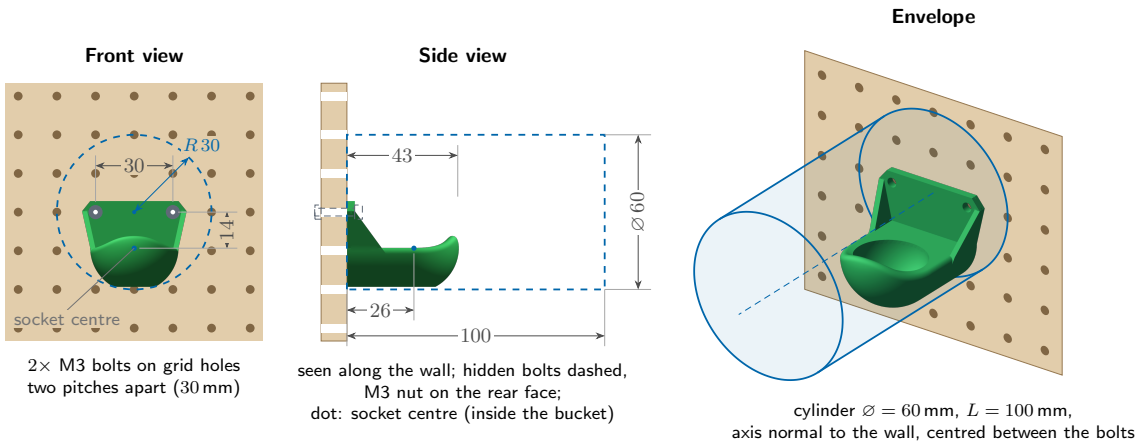


Figure 3: Hold specification: two M3 bolts on horizontally adjacent grid holes, passing through the hold and the board and tightened by nuts on the rear of the wall, and the cylindrical envelope ($\varnothing = 60\,\text{mm}$, $L = 100\,\text{mm}$) the hold must fit in. All dimensions in millimetres.

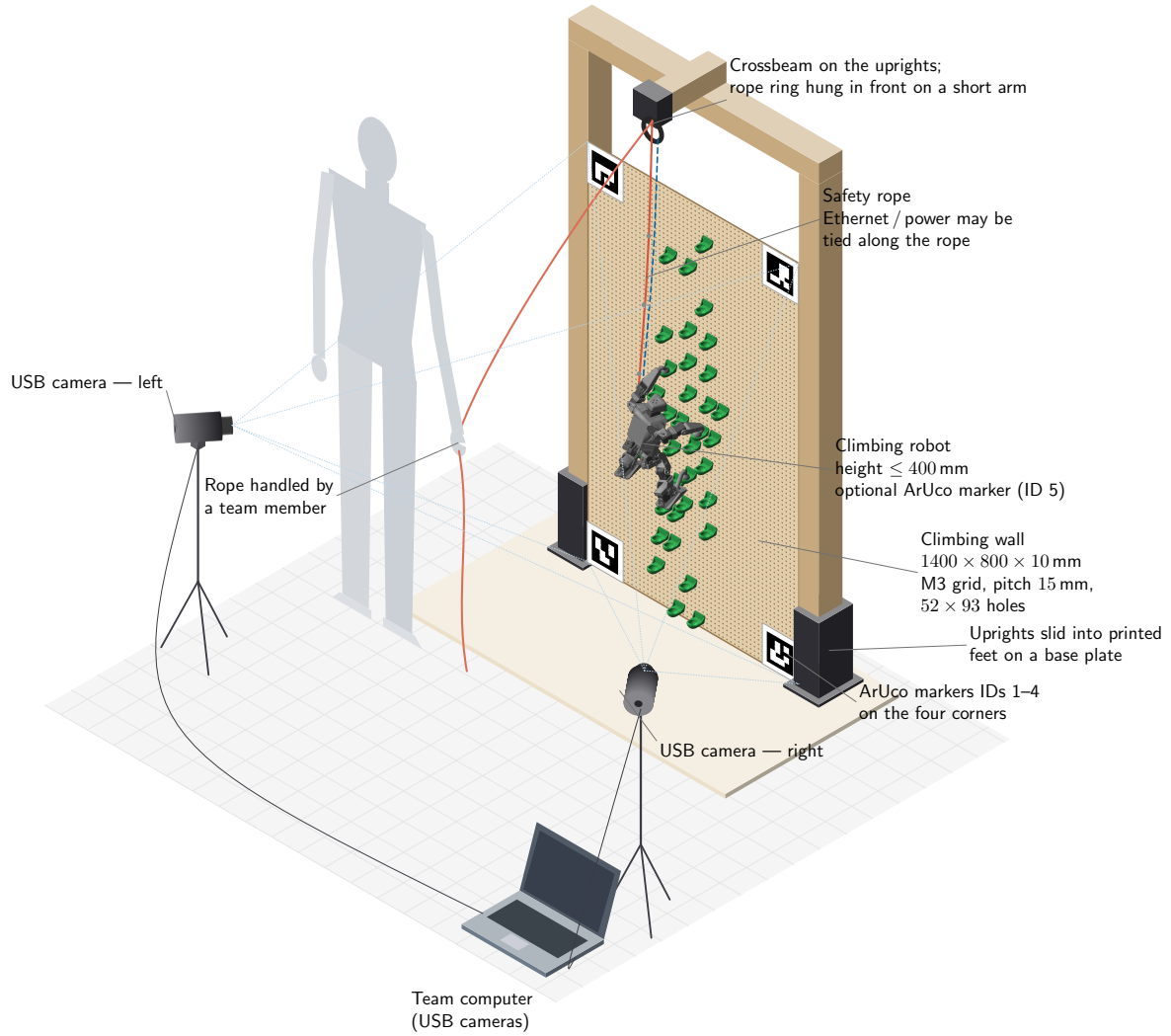


Figure 4: Isometric view of the setup: instrumented wall, crossbeam and pulley, safety rope with the optional Ethernet / power tether, and the two external USB cameras observing the wall.

5.2 ArUco markers

ArUco markers are provided by the organisers, ready to be attached to the **four corners** of the wall.

Each wall marker (Figure 5) is:

- a square **black code of** 100×100 **mm**,
- surrounded by a **white border of 18 mm**, i.e. 136×136 mm in total,
- placed so that the **outer corner of the white border coincides with the corner of the wall**.

The marker IDs are:

ID	Position
1	top-left corner of the wall
2	top-right corner of the wall
3	bottom-left corner of the wall
4	bottom-right corner of the wall
5	robot

The wall markers give the cameras their extrinsic parameters with respect to the wall frame, and therefore a common reference frame for all teams.

A team may attach **marker ID 5 on its robot**, **wherever it wants** on the robot and with the **size of its choice**, to be tracked by the external cameras.

All markers belong to the ArUco dictionary `DICT_4X4_50`.

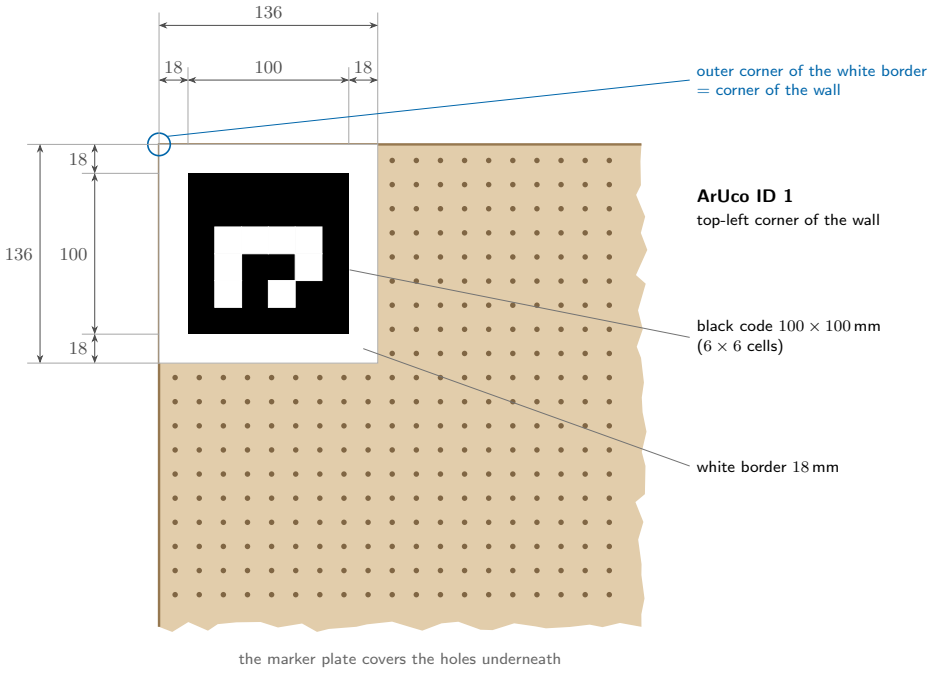


Figure 5: ArUco marker ID 1 at the top-left corner of the wall: 100×100 mm black code, 18 mm white border, outer corner of the border on the wall corner. The other corners are symmetric. Dimensions in millimetres.

5.3 Tether: Ethernet and power

Ethernet and power cables may be attached along the safety rope. They must follow the rope and must not be routed to a separate anchor point.

Cable strain relief must be provided on the robot side so that a fall loads the anchor point, never the connectors.

5.4 Anchor point

Each team must provide on its robot an **anchor point for the safety rope**.

6 Perception

Each run is performed in one of two perception modes. The mode only depends on whether ArUco markers are used; it determines the time factor applied in the ranking (Section 8).

6.1 With ArUco markers

The ArUco markers are attached to the four corners of the wall (Section 5.2). They give the **extrinsic parameters of the external cameras**, i.e. the pose of each camera with respect to the wall.

The robot may then be:

- **tracked with its own ArUco marker** (ID 5), attached by the team (Section 5.2): the robot pose in the wall frame is given directly by marker detection;
- or **detected without a marker on the robot**, e.g. by a learned model, the wall markers still providing the camera extrinsics.

6.2 Markerless

No ArUco marker is present, neither on the wall nor on the robot. The wall, the holds and the robot are perceived *naturally* from the camera images.

The camera extrinsics are not given. At the start of each attempt, the organisers **slightly move the external cameras**; estimating the pose of the cameras with respect to the wall is the job of the team's code.

6.3 On-board vision

The robot may also use its own on-board camera(s), alone or together with the external cameras. The same two modes apply: the run is *with ArUco markers* if markers are used, *markerless* otherwise.

7 Simulation

In addition to the real robot, teams are evaluated in simulation, using the **MuJoCo** physics simulator.

A simulator will be provided by the organisers at a later date. It is based on **MuJoCo (CPU)** and includes the models of the **robot**, the **walls** and the **holds**. It will be available at:

`https://github.com/rock-climbing-robots/simulator`

- The simulated competition uses walls of the **same three difficulty levels** as the real one.
- **Perception is simulated by MuJoCo**: the robot relies on the camera images rendered by the simulator. The two perception modes of Section 6 (with ArUco markers, markerless) apply in the same way, with the same time factors.
- Runs in simulation are ranked with the same rules as on the real robot (Section 8).

8 Evaluation and scoring

There are **two completely independent evaluations**, each with its own ranking: **in simulation** and **on the real robot**. Both use walls of three difficulty levels. The goal is to **climb the wall in the shortest time**.

Start of a run. The robot is either placed *on the ground*, anywhere around the wall, and gets onto the wall by itself; or placed *at the bottom of the wall*.

Height climbed. The height climbed is given by the **highest hold touched** by the robot. **Reaching the top** of the wall means touching one of the **highest holds**.

Ranking. Runs are compared by:

1. **height climbed** — higher always beats lower, whatever the time;
2. **time**, multiplied by the factors below.

Condition	Time factor
Wall difficulty level 3	×1
Wall difficulty level 2	×1.25
Wall difficulty level 1	×1.5
Start on the ground, robot gets onto the wall by itself	×1
Start at the bottom of the wall	×1.1
Markerless perception	×1
Perception using ArUco markers	×1.25
Humanoid robot	×1
Non-humanoid robot	×1.1

Factors multiply: for instance, a run on a level-2 wall, starting at the bottom of the wall and using markers, has its time multiplied by $1.25 \times 1.1 \times 1.25 = 1.71875$.

Timer. If the robot starts on the ground and gets onto the wall by itself, the timer **starts when its contact with the ground is broken**; if it starts at the bottom of the wall, the timer **starts at the very beginning** of the run. The timer **stops when the robot touches the very last hold**, or when the run fails — a partial success, ranked on the height climbed.